Partial Fractions

Questions 1–10

Question 1. Express $\dfrac{2}{x(x^2+2)}$ in the form $\dfrac{A}{x}+\dfrac{Bx+C}{x^2+2}$.

Question 2. Express $\dfrac{3x^2-x+10}{(x+1)(x^2+1)}$ in the form $\dfrac{A}{x+1}+\dfrac{Bx+C}{x^2+1}$.

Question 3. A student is trying to express a rational function in partial fractions. Their working is shown below:

$$\frac{3x^2+5x+5}{(x+1)(x^2+2)} = \frac{A}{x+1} + \frac{B}{x^2+2}.$$

a) Explain the mistake made by the student.

b) Express $\dfrac{3x^2+5x+5}{(x+1)(x^2+2)}$ correctly as partial fractions.

Question 4. Write each of the following as partial fractions:

a) $\dfrac{x+5}{(x^2+1)(x+1)}$

b) $\dfrac{5x}{(2x+1)(x^2+1)}$

c) $\dfrac{x-2}{(x+3)(x^2+4)}$

d) $\dfrac{3x^2-6x+15}{(2x^2+1)(x-1)}$

e) $\dfrac{25x^2}{2x^3-3x^2+8x-12}$

f) $\dfrac{x^2}{x^3-3x^2+5x-15}$

g) $\dfrac{1}{(x^2+k)(x-1)}$, where k is a positive constant.

h) $\dfrac{k}{(x^2+k)(x-k)}$, where k is a positive constant.

Question 5. It is known that

$$\frac{6x^2+5x+6}{x^3+kx} \equiv \frac{A}{x} + \frac{Bx+C}{x^2+k},$$

where k, A, B and C are positive integers. Determine the possible values of k. **The following questions require the use of Partial Fractions and A-Level Maths Integration Methods.**

Question 6. Find each of the following in simplified form:

a) $\displaystyle\int \frac{18}{(x-1)(x+2)(x-4)}\,dx$

b) $\displaystyle\int \frac{x-2}{(x-1)^2}\,dx$

c) $\displaystyle\int \frac{7x}{(2x-1)(x+3)^2}\,dx$

Question 7. Show that

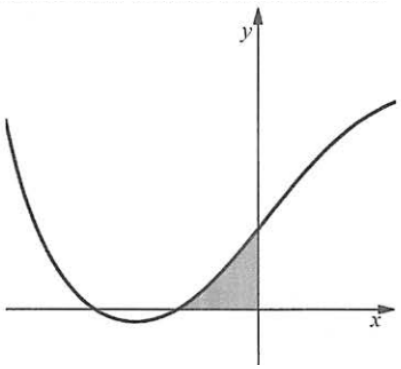
$$\int \frac{3x^2 + x + 4}{(x+1)(x^2+2)}\,dx = \ln\left((x+1)^2\sqrt{x^2+2}\right) + c.$$

Question 8. Show that

$$\int_1^4 \frac{x^2 - 4x + 15}{(x+1)(x^2+3)}\,dx = \ln\frac{3125}{722}.$$

Question 9. The diagram below shows part of the curve with equation

$$y = \frac{4x^2 + 6x + 2}{(x+2)(x^2+2)}, \qquad x \neq -2.$$



Find the exact value of the shaded area.

Question 10. Find the value of k such that

$$\int_4^6 \frac{(k-1)x^2 - 3kx - 4}{(x-3)(x^2+4)}\,dx = \ln\left(\frac{32}{3}\right).$$